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Carbon accounting quality: Measurement and the role of assurance☆

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ABSTRACT

We examine whether external assurance improves the quality of firms' carbon accounting. We develop a measure of carbon accounting quality based on the deviation of reported emissions from a model-based expected level and supplement it with two survey-based measures. We show that assurance is associated with higher carbon accounting quality. This association is stronger when firms have weaker pre-existing carbon accounting systems and when assurance is more thorough. Consistent with assurance enhancing quality, assurance relates to the identification of issues in a firm's carbon accounting system, along with fewer omissions and more revisions of prior disclosures. Exploiting mandated assurance for non-financial reporting in three E.U. countries, we show that firms experience post-regulation increases in carbon accounting quality. Together, the findings highlight the importance of assurance for reporting firms that are improving their carbon accounting quality.

1. Introduction

Combating climate change is an essential theme around the world, with more than 100 countries—or nearly 50 % of world

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GDP—having net-zero carbon commitments (ECIU, 2024). Achieving these goals requires organizations to accurately quantify carbon footprints, which gives rise to carbon accounting. In the corporate sector, which released an estimated 49 % of emissions in the U.S. in 2021 (EPA, 2023), the strong demand for high-quality emissions data from shareholders and other stakeholders has resulted in significant increases in emissions disclosure (Cohen et al., 2023b; Dikolli et al., 2022). This demand reflects the decision-usefulness of emissions data, which research shows influences resource and risk allocations in the economy (e.g., Bolton and Kacperczyk, 2021, 2023; Ilhan et al., 2023). New regulations—from the E.U. to California and China—further underscore the push for credible emissions reporting. However, more disclosure does not guarantee accuracy. It is therefore critical to evaluate the quality of firms' reported emissions and understand how firms can improve their carbon accounting quality.

This study develops a measure of carbon accounting quality and examines the role of assurance, i.e., third-party verification, in shaping carbon accounting quality. We define carbon accounting quality as the extent to which reported emissions reflect an entity's actual emissions. There are two reasons why a carbon accounting system would not perfectly report an entity's actual emissions. First, the *measurement standard*, such as the Greenhouse Gas Protocol ("GHG Protocol"), may be flawed (Kaplan and Ramanna, 2021; Glenk, 2025; Reichelstein, 2024). The GHG Protocol, a measurement standard voluntarily followed by most firms globally, lays out rules for tracking and reporting emissions. Second, a carbon accounting system inherently involves estimations and discretion, and thus has the potential for errors and biases, i.e., *implementation issues*. For example, a firm may not have proper metering equipment in all its facilities to compute its emissions. Because the use of the GHG Protocol is so pervasive across firms, any broad sample archival assessment of carbon accounting quality can only be performed within this measurement standard. Therefore, we focus our predictions and inferences on implementation issues.

We hypothesize that third-party verification ("assurance") can improve a firm's carbon accounting quality primarily by identifying the firm's implementation issues. The assurance process includes procedures such as checking samples of reported data against source data, interviewing relevant personnel, reviewing internal and external documentary evidence, and conducting on-site visits. Importantly, assurors can engage in "process assurance" which involves reviewing the reporting firm's information systems for collection, aggregation, analysis, and internal verification and review of environmental data. To the extent implementation issues are discovered through assurance and subsequently resolved by reporting firms, we should expect an improvement in carbon accounting quality post-assurance.

Financial statement audits improve (at least the perceptions of) financial reporting quality; however, it is unclear whether this inference generalizes to assurance of carbon accounting. First, mandatory financial statement audits have existed for almost 100 years in the U.S. with highly standardized rules, procedures, and outcomes. These audits cover firms' accounting, where standards and common practices date back several hundred years. By contrast, carbon reporting practices, measurement standards, and assurance are all evolving at the same time and arguably are still in very early stages, with the reporting standard (i.e., the GHG Protocol) only developed in the 1990s. As such, firms are still learning to properly measure emissions. Second, financial audits primarily address agency frictions due to the separation of ownership and control (Watts and Zimmerman, 1983). ESG reporting targets a much broader stakeholder base, and agency theories in this area remain underdeveloped. Voluntary carbon assurance may primarily address measurement and reporting issues because the practice is in its infancy and may not necessarily address agency issues, though its efficacy in addressing either is unknown. Third, the accounting for—and, therefore, assurance of—carbon emissions is fundamentally different, e.g., lacking a balance sheet equivalent (e.g., Reichelstein, 2024) or a mechanism for long-run true-ups, like accruals being settled via cash flows. Practitioners emphasize different assurance procedures, such as site visits and interviews, as opposed to data analytics and testing.[1] Because of these institutional differences, it is unclear whether external assurance improves carbon accounting quality, let alone *how much*. In addition, with decarbonization and combating climate change being important global themes, voluntary carbon assurance and regulations that involve mandatory assurance are both becoming increasingly common. Understanding whether and to what extent assurance—even if it is voluntary—affects carbon accounting has the potential to inform practitioners and regulators.

The GHG Protocol groups emissions from a firm's operations and activities into three types (or "scopes"): direct emissions from controlled assets (Scope 1), indirect emissions from energy consumption (Scope 2), and other indirect emissions along the value chain, e.g., by suppliers and customers (Scope 3). We focus on Scopes 1 and 2 emissions because those emissions are within a firm's control to measure and report. We develop a measure of carbon accounting quality at the firm-year-scope level through a two-step process. First, we model reported emissions levels as a function of firm fundamentals that capture production activities and technologies. Second, we define "abnormal emissions" as the absolute value of the difference between reported and modeled emissions levels, i.e., the residual term from the regression. The intuition is that emissions are a byproduct of a firm's operations. Typically, the more a company produces and sells, the more greenhouse gases the firm directly emits (i.e., Scope 1) and the more energy the firm uses and, therefore, the more greenhouse gases it indirectly emits (i.e., Scope 2). Deviations from expected levels are therefore likely to reflect emissions measurement implementation issues.

We pair this measure of carbon accounting quality with assurance data at the scope level. Our assurance data come primarily from

[1] Further, there exists heterogeneity in the assurance services provided, with more than 90 % of the assurance cases being limited assurance, which entails substantially less assuror work than reasonable assurance, the level required for financial statement audits. Anecdotally, some assurors are willing to provide limited assurance for a fee as low as $15,000—roughly 0.7 % of the financial audit fee for an average U.S. public firm in 2021—and simply check spreadsheets shared by the reporting firm. Besides, because obtaining assurance allows firms to obtain higher ratings from some ESG raters, reporting firms may have the incentive to simply "check the box" and do the bare minimum—treating assurance as a mere formality.

firm disclosures in ESG reports, which are manually collected by Gipper et al. (2025). We supplement this data with assurance information from CDP, the world's largest platform of voluntary environmental disclosures. Through questionnaires, CDP requests extensive climate-related information (including emissions and assurance) from a broad set of global firms every year (Cohen et al., 2023b). Incorporating CDP data allows us to identify assurance for firms that do not issue ESG reports. Our final sample includes 998 U. S. firms with carbon emissions disclosure from 2010 to 2020. On average, 48 % of the firm-years report assurance over at least one emissions scope. In terms of the quality of carbon emissions reporting, firms' reported emissions differ from model-estimated emissions by 32 % on average (18 % at the median). The median firm, with approximately 206 thousand tons of Scope 1 emissions, deviates from our model by 37 thousand tons, about equivalent to all emissions from 371 flights between Los Angeles and New York using a narrow-body commercial jetliner.

In our main analysis, we document a strong negative association between assurance and abnormal emissions for both Scopes 1 and 2 emissions. We triangulate our findings using two alternative measures of carbon accounting quality from the CDP questionnaires, including management uncertainty in emissions reporting and the time lag between questionnaire release and completion. Both alternative measures negatively correlate with assurance. For all measures, our findings are consistent with the prediction that assurance improves carbon accounting quality.

We perform a number of cross-sectional tests to strengthen our inference. Importantly, our result is stronger for firms with *ex ante* weaker carbon accounting systems, i.e., firms not subject to alternative reporting regimes that could improve carbon accounting even without assurance. For example, for firms subject to the U.S. Environmental Protection Agency's (EPA) Greenhouse Gas Reporting Program (GHGRP), which requires high-emitting facilities to report Scope 1 but not Scope 2 emissions, our finding is weaker for Scope 1 but not for Scope 2 emissions. This asymmetry not only lends credibility to our hypothesis but also provides additional validation for our emissions quality measure. Additionally, our result is stronger when assurance is more thorough, particularly when it covers all scopes of emissions or is conducted at the "reasonable" level used in financial audits, rather than at the "limited" level that is much more common in sustainability assurance. Overall, these findings support our interpretation of the main result that assurance improves carbon accounting quality.

We next explore the mechanism through which assurance could improve carbon accounting quality, documenting actions that assurors take along the causal chain. We use CDP questionnaire data to develop three specific indicators of how carbon accounting systems function and correlate these measures with assurance. We show results that are consistent with assurance improving carbon accounting quality primarily by helping reporting firms identify issues in their carbon accounting systems, which subsequently results in fewer sources of emissions being omitted and historically reported emissions figures being updated via revisions.

Thus far, our analysis focuses on voluntary assurance in the U.S. While our supplementary tests support a causal interpretation, we cannot completely rule out endogeneity concerns. Moreover, even if voluntary assurance does causally improve carbon accounting quality, it is not clear that the insight generalizes to an assurance mandate (e.g., Breuer, 2021). In our last analysis, we use the implementation of the E.U.'s Non-Financial Reporting Directive (NFRD) as a shock to assurance. Adopted by the E.U. in 2014 and fully implemented close to 2018, NFRD requires certain public companies to provide non-financial disclosure documents. Although NFRD is a disclosure mandate, three member countries also opted for mandatory assurance over reporting firms' sustainability disclosures, creating variation within the E.U. regarding mandated assurance of required emissions disclosures. Using a difference-in-differences design, we show that firms in countries with assurance mandates experienced an increase in carbon accounting quality after NFRD implementation relative to other E.U. firms. This evidence suggests that the benefits of assurance extend beyond voluntary adoption, supporting a causal interpretation and highlighting the potential effectiveness of assurance mandates.

Our study makes three contributions. First, we contribute to the ESG reporting literature, and in particular, the quality of sustainability disclosure. Non-financial reporting began outside financial reports as a voluntary response to stakeholder demands. However, such reporting has increasingly been mandated through regulation (e.g., Gibbons, 2024; Krueger et al., 2024) and integrated into mainstream communication channels such as conference calls, annual reports, and even financial statements (e.g., Sautner et al., 2023; Agrawal et al., 2024; Muller et al., 2025). This evolution has fueled research on the quality of ESG disclosures, which has traditionally focused on ESG topics broadly, emphasized disclosure breadth, and often relied on vendor-created scores.[2] For emissions reporting, recent work has taken a normative approach by studying the design of carbon accounting rules (e.g., Reichelstein, 2024; Glenk, 2025; Penman, 2025). Our innovation is to develop a measure of carbon accounting quality that evaluates reported emissions rather than the volume of disclosure, is independent of data vendor coverage, and is specific to each emissions scope. Because carbon emissions are central to assessing firms' exposure to carbon-related risks and thus affect capital allocation (e.g., Bolton and Kacperczyk, 2021), our measure provides investors, regulators, and other stakeholders with a tool for more informed decision-making.

Second, we contribute to the literature on the emergence and voluntary adoption of verification services, including auditing and new forms of assurance of non-financial information or processes (e.g., Watts and Zimmerman, 1983; Minnis, 2011; Duflo et al., 2013; Bourveau et al., 2024, 2025a; Schoenfeld, 2024; Liu and Tong, 2025). Our study is particularly related to ESG assurance, of which the U.S. has seen a striking increase over the last decade. Because ESG assurance is voluntary, lacks regulatory oversight, is primarily performed at a "limited" level as opposed to reasonable, and is highly heterogeneous in many dimensions, it is unclear whether such assurance is useful for firms and stakeholders (Gipper et al., 2025). We focus on assurance over emissions metrics and demonstrate its

[2] For example, prior papers have used disclosure scores from Bloomberg and Refinitiv (Lopez-de-Silanes et al., 2020), restatement of metrics in ESG reports (e.g., Ballou et al., 2018; Michelon et al., 2019), integrated reporting ratings from practitioners such as Ernst & Young (Barth et al., 2017; Maroun, 2019), and disclosure indices developed by researchers (Plumlee et al., 2015). Those measures typically focus on the *amount* of disclosure (e.g., the number of ESG metrics disclosed) and consider both *qualitative* and quantitative disclosures.

usefulness in improving carbon accounting quality.[3]

Third, our study is related to a long literature that examines the economic consequences of financial statement audits. Prior studies examine the rigor of audit (e.g., SOX provisions), enforcement (e.g., PCAOB inspections), and industrial organization (e.g., the fall of Arthur Andersen).[4] Because audits of public firms are mandatory, those studies focus on variation within mandatory audit regimes rather than audit adoptions. Studies that provide direct evidence of the audit itself are not common (DeFond and Zhang, 2014, p. 292). A handful of studies examine what happens from non-audit to audit for private firms but do not focus on reporting quality (e.g., Blackwell et al., 1998; Allee and Yohn, 2009; Kim et al., 2011; Breuer, 2021; Bourveau et al., 2025a). One exception is Minnis (2011), who shows that accruals are better predictors of future cash flows for private firms with a financial statement audit. Therefore, our study contributes to the broader audit literature by directly examining the reporting consequences of *audit adoptions*.

2. Conceptual underpinning

2.1. Reporting emissions

Emissions data are most often measured and reported to follow the Greenhouse Gas (GHG) Protocol. The GHG Protocol was developed in the 1990s around the signing of the Kyoto Protocol, a U.N.-organized emissions reduction plan and predecessor to the 2016 Paris Agreement. As (i) countries started to commit to global decarbonization and (ii) there were no guidelines for organizations to measure emissions, the World Resources Institute (WRI, a sustainability-focused not-for-profit organization) and the World Business Council for Sustainable Development (WBCSD, a consortium of global firms) responded by creating the GHG Protocol. Over time, WRI and WBCSD have created and updated standards for corporations, cities, and products, among others. The standards most applicable to corporations are the GHG Protocol Corporate Accounting and Reporting Standard and the Corporate Value Chain (Scope 3) Standard. Following convention, we refer to those two standards as the GHG Protocol in this study.

The GHG Protocol distinguishes between three emissions sources labeled Scopes 1–3. Scope 1 emissions are "direct" emissions from sources that are controlled or owned by a firm. Scope 2 and Scope 3 emissions are "indirect"—meaning they are not produced by the company itself. Scope 2 emissions are associated with the purchase of electricity, steam, heat, or cooling. Scope 3 emissions cover the entire value chain from sources not controlled by the company, such as emissions due to the use of the firm's products or employees' work-related travel.

The procedures of the corporate GHG Protocol can be summarized in three steps. The first step is to choose the organizational boundary—either equity share or control—for consolidating GHG emissions. This step is somewhat analogous to the concept of consolidation used in financial reporting in determining what to include or not include in financial statements. The second step is to choose the operational boundaries, i.e., how emissions can be classified into Scopes 1, 2, and 3. The third step is to calculate GHG emissions. The common procedure is to identify all emissions sources within the chosen boundaries, multiply a measure of activity by a corresponding emissions conversion factor for each emissions source, and aggregate the estimated emissions. The common practice is to express emissions in carbon dioxide equivalents (CO_2e), meaning that non-CO_2 greenhouse gas emissions (e.g., methane) are converted to CO_2e using the appropriate conversion factors, thereby normalizing emissions to the most prevalent greenhouse gas, CO_2.

2.2. Conceptualizing carbon accounting quality

We conceptualize carbon accounting quality as the extent to which reported emissions reflect an entity's actual level of emissions.[5] Our conceptualization of carbon accounting quality mirrors the framework proposed by Dechow et al. (2010) to analyze earnings quality. Dechow et al. (2010) argue that reported earnings is a *function* of an enterprise's financial performance during a reporting period. Earnings quality is an evaluation of this function, which is an accounting measurement system that establishes a mapping between reported earnings and financial performance. There are three reasons why reported earnings would not perfectly measure financial performance. The first is multiple decision models, i.e., the ability of earnings to summarize a firm's unobservable financial performance for decision-making. An accounting system that produces a single reported earnings number cannot properly represent the underlying financial performance that is relevant for all decision-makers. The second is measurement standards. No standard can perfectly measure performance for any given firm. The third is the implementation of an accounting system, which inherently involves estimations and discretion.

[3] Our study is related to papers that study the role of assurance in ESG reporting quality more broadly. Using scores assigned by Ernst & Young, Maroun (2019) shows that external assurance increases the quality of integrated reports—which include both financial and other value-relevant information—in South Africa. Ballou et al. (2018) and Michelon et al. (2019) document a positive association between assurance in ESG reports and the probability of a reporting firm revising its reported numbers—regardless of metric type—in subsequent years. Luo et al. (2023) find a positive association between carbon assurance and CDP disclosure scores. However, assurance has been explicitly incorporated into CDP ratings since 2017, so a positive association between assurance and CDP ratings would likely be mechanical.

[4] For studies exploiting SOX, see, e.g., Doyle et al. (2007), Ge and McVay (2005), Ge et al. (2017), and Armstrong et al. (2024). For studies exploiting PCAOB inspections, see, e.g., DeFond and Lennox (2017), Shroff (2020), Gipper et al. (2020), and Stuber and Hogan (2021). For studies exploiting the fall of Arthur Anderson, see, e.g., Guo et al. (2022).

[5] We adopt phrasing where emissions are converted to carbon equivalents in practice, i.e., normalizing any air-based emissions to the prevalent and impactful greenhouse gas carbon dioxide.

These three factors from Dechow et al. (2010), which influence earnings quality, can similarly apply to carbon accounting quality. First, in the context of carbon accounting, we believe that actual emissions track a firm's carbon footprint in a straightforward way, but reported emissions produced by the carbon accounting system may not be equally relevant in all decisions. Dechow et al. (2010) note that an accounting system would not perfectly measure financial performance due to many decision models that could utilize the measure. For example, earnings quality could be low for stewardship purposes (e.g., as a bonus plan metric) but high for valuation purposes (e.g., as an input to an investor's enterprise value calculation). Similarly, a carbon accounting system could generate a measure that would be an input to multiple decisions, such as meeting a "net zero" commitment or quantifying the exposure to regulatory risk arising from cap and trade or a governmental entity's emissions reduction policies.

Second, the measurement standards (e.g., the GHG Protocol) could be flawed. For example, a widely discussed case is the GHG Protocol's Scope 3 emissions, which are emissions measured along the value chain. Research in measurement standards focuses predominantly on Scope 3 emissions. For example, Kaplan and Ramanna (2021) contend that "the same emissions are reported multiple times by different companies, while some entities entirely ignore emissions from their supply and distribution chains." In addition, Glenk (2025) argues that the GHG Protocol fails to represent a firm's emissions faithfully.

Third, a carbon accounting system inherently involves estimations and discretion, and thus has the potential for errors and biases, i.e., implementation issues. These issues could arise from data management flaws, data gaps, unreasonable assumptions and extrapolations, and improper sampling, among others. For example, many firms have historically relied on Excel spreadsheets as opposed to carbon accounting software to track emissions. Errors could arise from data entry, aggregation, incorrect Excel formulas, etc. As another example, some firms do not have information systems in place to gather emissions data from necessary sources, such as energy bills for leased facilities. CDP classifies implementation issues into several non-mutually exclusive groupings. We use these groups to illustrate those issues and provide anecdotes in Appendix B.

Implementation issues could result in either under- or over-stating a firm's carbon emissions. For example, emissions will be underreported when data gaps exist. Overreporting is also plausible. For example, emissions would be overstated if a firm uses the utility bills from the fourth quarter—during which the firm may use more energy than average due to seasonality—to extrapolate emissions for the entire year. More generally, because carbon emissions measurement is still in its infancy, a carbon reporting group within the firm may also be growing and still developing the knowledge and expertise to estimate emissions correctly.[6]

The above-discussed implementation issues focus on errors and abstract away from managerial incentives to manipulate emissions. This contrasts with the earnings quality literature, in which implementation issues include both innocuous errors and manipulations by managers. Manipulation could theoretically exist. Such incentives could stem from CEO compensation being tied to emissions reduction goals. For example, because emissions are aggregated from all sources within the reporting boundary and firms have the discretion to define their reporting boundary, a firm could strategically choose a reporting boundary that results in lower reported emissions.

2.3. *Assurance and carbon accounting quality*

By assurance, we refer to third-party verification of firm-disclosed ESG metrics, specifically GHG emissions. Assurance over emissions resembles a financial statement audit but differs in many dimensions (e.g., Gipper et al., 2025). In general, ESG assurance is voluntary in most countries, including the U.S. The assurance providers include both traditional auditors who also perform financial statement audits and ESG engineering and consulting firms with specialties in environmental and social matters, i.e., non-financial assurors. In the U.S., roughly 80 % of assurance services are provided by non-financial assurors. In terms of assurance quality, the contracted level of assurance is mostly "limited assurance," which is somewhat analogous to a "review" engagement for quarterly financial reports in the U.S. Reasonable assurance, which is the level of assurance required for annual financial statement audits, is very uncommon in the non-financial space. Assurance providers use a variety of assurance standards. Both financial and non-financial assurors offer separate consulting and advisory services related to non-financial reporting. In the aggregate, although assurance over ESG metrics is increasingly common, there exists considerable heterogeneity in many dimensions compared to financial audits.

Our framework of carbon accounting quality entails multiple decision models, measurement standards, and implementation issues. However, firms' intent regarding assurance appears to follow—rather than precede or influence—firms' and stakeholders' decision models around emissions and choice of measurement standards. For example, the GHG Protocol is pervasively used; thus, an archival assessment of carbon accounting quality can only be done within this measurement standard. Therefore, we hypothesize that assurance can increase a firm's carbon accounting quality by identifying implementation issues in reporting firms' carbon accounting systems. Although there is significant heterogeneity in what an assurance engagement entails, the third-party assurance process could include procedures such as checking samples of reported data against source data, interviewing relevant personnel, reviewing internal and external documentary evidence, and conducting on-site visits. Assurors also engage in "process assurance" which involves

[6] For example, Enel S.p.A. disclosed emissions that were four times global total emissions in 2009 (Guardian, 2011).

reviewing the reporting firm's information systems for collection, aggregation, analysis, and internal verification and review of environmental data.[7] To the extent implementation issues are discovered through assurance and subsequently resolved by reporting firms, we should expect an improvement in carbon accounting quality post-assurance.

As discussed in the previous section, implementation issues could entail both errors and intentional misreporting. Our hypothesis is primarily based on assurors identifying implementation issues that do not necessarily involve managerial misreporting incentives. This is because the literature is nascent regarding evidence of managers manipulating quantitative emissions reports, though qualitative greenwashing appears rampant. Even in the earnings quality literature, the evidence of the relationship between managerial incentives and financial reporting quality is mixed (see, e.g., Armstrong et al., 2013). In addition, misreporting stems from incentives, which typically come from executive compensation contracts. As shown in Cohen et al. (2023a), only 1 % of the U.S. firms link compensation to carbon emissions. Therefore, we leave it to future research to explore whether managerial incentives may drive manipulated reporting of emissions and whether the effect of assurance on carbon emissions is partially driven by these managerial incentives.

Because our hypothesis does not incorporate intentional misreporting, a related issue is the necessity of repeated assurances. Repeated financial statement audits are often utilized to address ongoing agency frictions and managerial incentive problems (Watts and Zimmerman, 1983). We argue that repeated assurances have incremental benefits. For one, carbon accounting is in early stages and evolving, and firms are constantly learning about scoping and measuring carbon emissions sources. It is unlikely that firms adequately address issues uncovered through carbon assurance in a single period.[8] In addition, firms change operational practices regularly, including growing and shrinking their operations, installing new (perhaps more efficient) equipment and systems, etc. Additionally, even in financial statement audits, auditors not only uncover misrepresentations but also innocuous errors, which comprise a large portion of accounting restatements (Hennes et al., 2008). It is also not uncommon for firms to have repeated financial accounting errors. As such, we believe that repeated external assurance has benefits.

3. Empirical strategy

3.1. Research design

To investigate the association between assurance and carbon accounting quality, we estimate the following linear regression model:

$$CAQ_{i,t,s} = \beta\, Assurance_{i,t,s} + \text{Controls} + \text{Fixed Effects} + \varepsilon_{i,t,s} \tag{1}$$

In Eq. (1), *CAQ* is one of the three measures of carbon accounting quality for firm *i*'s Scope *s* emissions in year *t*. We discuss these measures in Section 3.2. Our independent variable of interest, *Assurance*, is an indicator variable equal to one if a firm's carbon emissions for Scope *s* are assured in year *t*. Although there is no theoretical guidance for the selection of controls, we consider common firm characteristics, including firm size, ROA, book-to-market, leverage, and an indicator for loss firms (Gipper et al., 2025). The purpose of including these firm characteristics is to mitigate the concern that firm characteristics jointly determine assurance and carbon accounting quality. For example, larger firms may have the resources to set up a robust carbon accounting system and engage an assuror. Including firm size allows us to control for this possibility. Eq. (1) includes industry and year fixed effects. Year fixed effects remove time trends in carbon accounting quality. Inherently different production functions across industries could lead to heterogeneous measurement problems in emissions, resulting in differences in *CAQ*. Industry fixed effects remove the influence of these differences, along with other industry-invariant confounding factors. All continuous variables are winsorized at the 1 % and 99 % levels. Throughout the study, we base our inferences on standard errors clustered by firm.

In subsequent tests, we examine whether the association between assurance and carbon accounting quality varies in the cross section by partitioning the scope-specific *Assurance* indicator into non-overlapping variables as follows.

$$CAQ_{i,t,s} = \beta_1\, Assurance_{i,t,s} \times Partition_{High} + \beta_2\, Assurance_{i,t,s} \times Partition_{Low}\left(+ \beta_3\, Partition_{High}\right) + \text{Controls} + \text{Fixed Effects} + \varepsilon_{i,t,s} \tag{2}$$

The partitioning variables are binary indicators set to one for the group of firms with certain characteristics, and to zero otherwise. For example, when we investigate whether the association varies by whether the firm reports to the EPA, $Partition_{High}$ ($Partition_{Low}$) would equal one for EPA-reporting (non-EPA-reporting) firms. Together, $Assurance_{i,t,s} \times Partition_{High}$ and $Assurance_{i,t,s} \times Partition_{Low}$ span *Assurance*. β_1 and β_2 capture the total effect of assurance on carbon accounting quality for the EPA and non-EPA firms, respectively. We additionally include the split indicator, i.e., $Partition_{High}$, if the split is not solely defined for observations with assurance to

[7] For example, Apex Inc. provided assurance over Apple's Scopes 1 and 2 emissions. For the engagement, Apex undertook virtual remote site visits to Apple facilities in Israel and the U.S., conducted meetings with personnel from corporate offices, interviewed relevant personnel, reviewed internal and external documentary evidence, reviewed a sample of data against source data, and reviewed Apple's information system for collection, aggregation, analysis, and internal verification and review of environmental data. As another example, PwC provided assurance over Estée Lauder's Scopes 1 and 2 emissions, among others. PwC performed similar procedures, e.g., inquiries, tests of computations, reviews of supporting documentation, etc.

[8] For example, in its response to the CDP questionnaire, one company discusses how improving carbon accounting is a continuous process. "The first third party verification/assurance started a continuous improvement process that allowed improving the quality of the information gathered for the 2014, 2015 and 2016 carbon footprint."

control for the inherent difference in carbon accounting quality across partitioned firms (in this case, whether the firm reports to the EPA). This approach has been frequently used in prior studies (e.g., Christensen et al., 2016; Zhu, 2019; Jayaraman and Wu, 2019).[9] To statistically compare the heterogeneous effects, we perform a Wald test of coefficient equality for our partitioning variables.

3.2. Measures of carbon accounting quality

Our first measure is based on the deviation of reported emissions from a model-based expected level. This approach is rooted in the fundamental idea that carbon emissions are a byproduct of a firm's production process; therefore, we model emissions as a linear function of predictors that capture a firm's production as follows:

$$\mathrm{Ln}\left(Emissions_{i,t,s}\right) = \beta_{industry(i),s}\, Determinants_{i,t,s} + Year\ FE_{industry(i),t,s} + \varepsilon_{i,t,s} \tag{3}$$

where a firm i's Scope s emissions level in year t is modeled as a function of determinants (more below) and year effects estimated for Scopes 1 and 2 separately. The economics of production likely differ by industry, so we allow the coefficients for each determinant to vary by industry.[10]

Our first set of determinants aims to capture production output. These determinants include firm sales, changes in inventory, and cost of goods sold. Because a firm's inventory changes and sales reconcile to production, these determinants allow us to proxy for production output. Our second set of determinants intends to capture production technology. We include firm size and net property, plant, and equipment (PP&E) to proxy for the existence and scale of operating facilities, such as factories and machines that generate emissions (e.g., Nguyen et al., 2021). We also include fixed asset tangibility (net PP&E scaled by total assets) because firms with many fixed assets are likely to operate very differently compared to firms with many intangibles (Downar et al., 2021; Nguyen et al., 2021). In addition, we include capital expenditure and asset age (defined as accumulated depreciation divided by annual depreciation) because production technology evolves, with more recent fixed assets more likely to be energy efficient and emitting less (e.g., Whitaker et al., 2012). We further consider the number of countries and states with operations (Goldhammer et al., 2017). *Ceteris paribus*, we expect centrality of production to generate economies of scale in production and reduce emissions; centrality also decreases the need for intra-firm transportation, reducing emissions. Last, we include the number of employees. For new-economy firms, Scope 2 emissions come primarily from office space utilities, which are determined largely by employee base. Aside from production output and technology, we follow ESG data vendors (e.g., MSCI, 2016, Refinitiv, 2017) and include historical emissions.[11] To further account for technology, we include industry-year fixed effects.[12] We use the *absolute value* of the predicted residuals from this model of firm-year-scope-level emissions, i.e., the predicted value of ε or modeled versus reported emissions, as our measure of carbon accounting quality. We call this variable *Abnormal Emissions*.

A key advantage of our regression-based measure is that it can be estimated for any firm with disclosure of emissions. We triangulate our inference using two alternative measures of carbon accounting quality from CDP data. Formerly known as the Carbon Disclosure Project, CDP is the world's largest platform of voluntary environmental disclosure. Every year, CDP sends out questionnaires to a broad set of global firms that are constituents of prominent stock market indices, such as the S&P 500, asking for an extensive set of environment-related information, including carbon emissions (Cohen et al., 2023b). In some years of our sample period, CDP asks reporting firms to identify how uncertain they are about the reported emission figures, with options including below 2 %, 2–5 %, 5–10 %, 10–20 %, all the way up to 100 %. This measure captures a firm's *perception* of its carbon accounting quality. To the extent a manager is on average truth-telling, smaller managerial uncertainty should imply higher carbon accounting quality.

[9] Our analysis is equivalent to an interaction design, except that we directly estimate the total effect of assurance (as opposed to the incremental effect of assurance) for each group.

[10] We survey the literature and tabulate features of existing carbon emissions models in Tables IA–2 To evaluate the relevance of determinants nominated by prior studies, we regress emission levels on each determinant and report the regression R^2. Those models do not allow flexible coefficients for industry or year. In our setting, because carbon emissions disclosures are voluntary, estimating Eq. (3) for each industry-year pair will result in many industry-years with insufficient observations if requiring a minimum number of observations, as done in papers estimating abnormal accruals, especially in early years. We acknowledge this is a limitation. With the potential for mandatory disclosure in the future, subsequent research could introduce separate coefficients for each industry-year pair. In addition, our emissions model relies on data from financial statements. If the organizational boundary set by firms for emissions reporting differs from the way firms consolidate their financial statement numbers, our carbon accounting quality measure can be noisy. Given that firms do not often disclose their carbon reporting methodology, we cannot investigate this question empirically. As firms improve their disclosure over time, future research can investigate how often firms set organization boundaries and consolidate financial statements differently, and how to address this difference and improve our model.

[11] Because we do not estimate Eq. (3) for each industry-year pair, controlling for lagged emissions accounts for firms that may have different production technologies and, hence, emissions. Therefore, including this variable allows us to better isolate abnormal emissions. In addition, this approach aligns with the GHG Protocol's "consistency" principle, which emphasizes comparable and consistent reporting within-firm over time. At the same time, the residual term is then the amount of innovation in emissions levels that cannot be explained by a firm's production activities, meaning that the model would not isolate smooth, persistent carbon accounting issues. Hence, our results may underestimate the effect of assurance. Nonetheless, we estimate alternative versions of the model where we (1) drop lagged emissions and instead use changes in total emissions as the dependent variable, and (2) use both linear and machine learning models with and without lagged emissions to estimate emissions. Our inferences are similar. See Tables IA–3 and IA-4, respectively.

[12] In our model, we allow the coefficient of each determinant to vary by industry. To illustrate the relevance of determinants, in Tables IA–5, we present a version of our emissions model without industry-level coefficients.

In addition, a well-functioning carbon accounting system should be able to generate carbon disclosures more efficiently and timely. Prior studies use the time difference between the fiscal year-end date and the earnings announcement (or annual report filing) date as a measure of financial reporting system quality (e.g., Heitzman and Huang, 2019). Similarly, we use the difference between the CDP questionnaire release date and firm response submission date (*Filing Lag*) as an alternative measure of carbon accounting system efficiency. Anecdotes further justify the use of this measure (e.g., KPMG, 2024).

3.3. Data

To construct the sample, we combine carbon emissions data from Trucost, assurance data from Gipper et al. (2025) and CDP, and firm fundamentals from Compustat. Trucost carbon emissions data include both disclosed emissions from reporting firms as well as estimated emissions from non-disclosing firms. Because we are interested in the quality of firms' carbon emissions disclosures, we only consider firm-years with disclosed emissions. Data on assurance used in this study come from Gipper et al. (2025) and CDP. Gipper et al. (2025) compiled ESG reports issued by S&P 500 firms and collected information on various dimensions of assurance (e.g., assuror identity) at the metric level (hereafter, "GRS data"). We supplement GRS data with data from CDP, which we discussed in Section 3.2. Sometimes, firms report their GHG disclosure and assurance to CDP but do not issue standalone ESG reports. By combining GRS data with CDP data, we reduce the measurement error in reporting and assurance.

In our analyses, we use several samples. For our main analyses focusing on U.S. firms, our sample is a panel spanning 2010 to 2020, covering 998 firms. The number of observations varies for Scopes 1 and 2 emissions because not all firms disclosing Scope 1 emissions also disclose emissions from Scopes 2 and vice versa. We use this sample for our main, robustness, and cross-sectional analyses. This sample reflects the intersection of Compustat, CRSP, and Trucost with assurance coverage from GRS and/or CDP data. It includes all U.S. public firms with non-missing controls and data to estimate *Abnormal Emissions*. We also perform several tests using additional data from the CDP questionnaires, Refinitiv, and Bloomberg.

3.4. Descriptive statistics

Table 1 presents descriptive statistics. The firm-level rate of assurance is 48 %, slightly higher than Gipper et al. (2025). This is because the rate in Gipper et al. (2025) includes assurances in ESG reports for S&P 500 firms, whereas our data has a larger sample of firms and also includes emissions assurances reported via CDP. Of the 477 firms that obtain emissions assurance, 65 % of firms—i.e., 310 of 477—do not obtain assurance every year after the first instance, indicating that some managers perceive to gain sufficient benefits from only periodic emissions assurance.[13]

In terms of *Abnormal Emissions*, the average absolute value of the deviation between reported and model-predicted logged emissions is 0.32 and 0.31 for Scopes 1 and 2, suggesting that, on average, the reported emissions differ from model estimated emissions by roughly 30 %. For example, Archer-Daniels-Midland reported emissions of about 15,588 thousand tons of CO_2e Scope 1 emissions in 2014, according to Trucost. On the same scale, our model predicts emissions for this case to be 11,636, a difference in logs of 0.29. In addition, as we show in Fig. 1, there is significant heterogeneity in *Abnormal Emissions* across industries. Energy Equipment and Services have the highest average *Abnormal Emissions* in both Scopes 1 and 2. Banks and Food Products have the lowest *Abnormal Emissions* for Scope 1 and Scope 2, respectively.[14]

Table 1 also reports various indicators of the carbon accounting system, which we derive from the CDP questionnaire data. The sample size varies by variable because the CDP questionnaire varies by year, and some questions are only included in select years. In addition, these questions are conditional on firms disclosing an emissions number. Because firms are more likely to disclose Scope 1 emissions relative to Scope 2 emissions, we see larger sample sizes for Scope 1 emissions. On average, firms identify two categories of issues. The probability of omitting sources of emissions is 0.45 and 0.37 for Scopes 1 and 2, respectively, suggesting that reporting issues, such as underreporting via known omitted sources, are pervasive.

3.5. Validating abnormal emissions as a measure of carbon accounting quality

In addition to using alternative measures, we strengthen our inference by validating our *Abnormal Emissions* measure in several ways. First, a criticism of abnormal accruals models is that determinants only explain roughly 30 % of the total variation in working capital changes (e.g., McNichols, 2002). For carbon emissions, our determinants (untabulated) explain 96.2 % and 91.0 % of the variation in emission levels for Scopes 1 and 2, respectively. Given the limited residual variation, we believe the measurement error problem is not as severe as for accruals models.

Second, as firms gain more experience in reporting carbon emissions, we should expect improvements in carbon accounting quality within-firm over time, especially due to the increasing focus on emissions during the decade. We regress *Abnormal Emissions* on

[13] We caution that this share may be as low as 25 % (untabulated) and only appears higher in our data due to the final year of CDP available to us. Our CDP data ends with the 2020 questionnaire, often covering firms' 2019 fiscal year. In an untabulated analysis, our results are unaffected if we (i) drop 2020 or (ii) use 2020 CDP questionnaire data to infer assurance in firms' 2020 fiscal year (the approach we use to calculate the 25 %).

[14] We perform an ANOVA analysis by regressing *Abnormal Emissions* on various fixed effects to understand its sources of variation. As we show in Tables IA–6, industry-invariant factors only explain 2.4 % and 10.5 % of Scope 1 and Scope 2 variation, respectively; firm-invariant factors explain 35.4 % and 40.5 % of the variation.

Table 1
Descriptive Statistics
This table presents descriptive statistics. Panel A presents rates of assurance within the main sample and patterns of obtaining assurance conditional on assurance adoption. Panel B presents descriptive statistics of variables. Definitions of all variables are as described in Appendix A. All continuous variables are winsorized at the 1st and 99th percentiles.

Panel A. Assurance Frequency and Adoption

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Scope 1			Scope 2			Either Scope 1 or 2		
	Firm-Years	Firms		Firm-Years	Firms		Firm-Years	Firms	
		N	%		N	%		N	%
Firm-Years:									
With assurance	2643	474	49	2506	451	48	2695	477	48
Without assurance	2752	878	91	2696	859	91	2883	913	91
	5395	967	100	5202	944	100	5578	998	100
Firms with:									
At least one instance of assurance	3874			3633			3972		
Remove pre-assurance firm-years	(663)			(630)			(685)		
Assurance adoption sample:	3211	474	49	3003	451	48	3287	477	48
Assurance every year (in our data)	641	89	9	590	85	9	619	85	9
Assurance every year after adoption	425	82	8	448	80	8	448	82	8
Some breaks after adoption	2145	303	31	1965	286	30	2220	310	31

Panel B. Descriptive Statistics of Key Variables

		(1)	(2)	(3)	(4)	(5)	(6)
		N	Mean	SD	25th	Median	75th
Firm Characteristics							
Size		5395	9.83	1.48	8.79	9.77	10.87
ROA		5395	0.04	0.09	0.01	0.04	0.08
Book-to-Market		5395	0.57	0.56	0.23	0.42	0.73
Leverage		5395	3.04	5.66	0.99	1.71	3.20
Loss		5395	0.15	0.35	0	0	0
Meet or Beat		5371	0.04	0.19	0	0	0
Abnormal Accruals		4963	0.08	0.10	0.02	0.05	0.11
Carbon Emissions							
Assurance	Scope 1	5395	0.49	0.50	0	0	1
	Scope 2	5202	0.48	0.50	0	0	1
Abnormal Emissions	Scope 1	5395	0.32	0.45	0.08	0.18	0.36
	Scope 2	5202	0.31	0.47	0.08	0.17	0.34
Emissions Level	Scope 1	5395	5860	20,680	30	206	1610
	Scope 2	5202	1080	2740	75	256	911
CDP Responses							
Issues Identified	Scope 1	3517	1.99	1.31	1.00	2.00	3.00
	Scope 2	3454	1.91	1.31	1.00	2.00	3.00
Omissions	Scope 1	5215	0.45	0.50	0	0	1
	Scope 2	5215	0.37	0.48	0	0	1
Revisions	Scope 1	5215	0.05	0.22	0	0	0
	Scope 2	5215	0.03	0.18	0	0	0
Manager Uncertainty	Scope 1	3305	4.28	2.89	1	4	7
	Scope 2	3184	4.27	2.90	1	4	7
Filing Lag		4733	4.88	0.24	4.75	4.80	5.11
Scores from ESG Rating Agencies							
Refinitiv Disclosure Score		3692	71.57	29.52	79.18	82.57	86.24
Bloomberg E Disclosure Score		1765	33.21	15.94	21.49	34.71	45.54

indicators for each year, with 2010 as the base year and, hence, omitted. We include firm fixed effects to focus the analysis on the time series and mitigate the impact coming from the change in the mix of disclosing firms over the decade. Fig. 2 shows decreasing trends for both Scope 1 and Scope 2 *Abnormal Emissions*, indicating improved carbon accounting quality over time.

Third, we relate the abnormal emissions measure to other measures of ESG reporting quality in Table 2. We find a strong relation between *Abnormal Emissions* and the managerial uncertainty measure of carbon accounting quality based on CDP questionnaires (*Manager Uncertainty*) for both Scopes 1 and 2 emissions. Also, we relate *Abnormal Emissions* to disclosure scores from Bloomberg and Refinitiv. Refinitiv has a score that captures a firm's overall ESG reporting practice, and Bloomberg has a score that specifically captures a firm's environmental disclosures. For both rating agencies, a higher score indicates better disclosure practices. Because firms with high-quality carbon accounting likely have more robust ESG reporting, we expect and find *Abnormal Emissions* to be negatively correlated with ESG disclosure scores. Taken together, the findings provide us with confidence that our *Abnormal Emissions* measure reasonably captures carbon accounting quality.

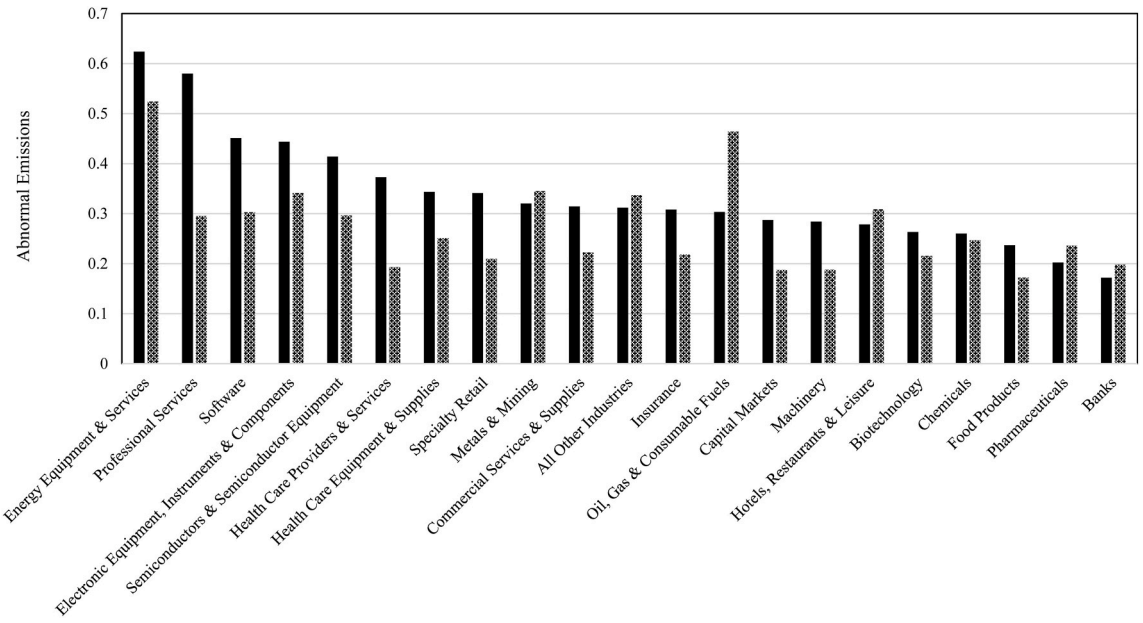


Fig. 1. Abnormal Emissions by Industry
This figure presents the sample average of *Abnormal Emissions* for various Trucost industry classifications, which has granularity similar to three-digit SIC (Bolton and Kacperczyk, 2021). We separately report scope-level averages for each of the 20 most populated industries and group the remaining industries into the "All Other Industries" category for parsimony. Bars filled in solid black (dots) depict results for Scope 1 (Scope 2). All calculations are based on the full sample of data at the firm-year level.

4. Results

4.1. Main result

We report our main findings in Table 3 Panel A. In columns (1) and (3), we report the univariate correlation between assurance and carbon accounting quality for Scopes 1 and 2, respectively. Note that both *Assurance* and *Abnormal Emissions* are measured at the firm-year-scope level. Because *Abnormal Emissions* inversely captures carbon accounting quality, the negative coefficients are consistent with third-party assurance *improving* carbon accounting quality. In columns (2) and (4), we include industry and year fixed effects, as well as common firm characteristics as control variables. Those controls and fixed effects partially mitigate the concern that firm and/or industry characteristics jointly determine improvement in carbon accounting quality and the decision to assure emissions metrics. For example, the coefficient on firm size is negative, suggesting that larger firms have higher carbon accounting quality. This finding is in line with larger firms having better financial reporting quality (Dechow et al., 2010). The coefficient on *Assurance* continues to be negatively significant. More importantly, the economic magnitudes are sizable. Assurance increases the quality of Scopes 1 and 2 emissions disclosure by 26 and 34 % relative to the sample mean, respectively.

In Panel B, we examine the robustness of our finding using two alternative measures of carbon accounting quality: managers' confidence in their carbon accounting systems (*Manager Uncertainty*) and the efficiency of the carbon accounting system (*Filing Lag*). The findings are similar. Together, the findings in Table 3 present strong evidence that assurance relates to carbon accounting quality and is consistent with assurance *increasing* carbon accounting quality.[15]

4.2. Addressing alternative explanations

Absent exogenous variation, our main finding could be subject to alternative explanations. One alternative is that the documented

[15] We perform a battery of robustness tests to strengthen our inference. To mitigate the concern that our finding is driven by the misspecification of the emissions model, we show that our findings are robust to many modifications, including (1) allowing each determinant of the emissions model to have separate coefficients for each year or each industry and half-decade combination (whereas in the baseline model, each determinant has a separate coefficient for each industry), (2) excluding firms in banking and utilities industries, (3) using additional firm-by-state level determinants, where available, and (4) creating a principal component measure from the three measures in both Panels A and B. See Tables IA–7 In addition, we perform several tests to address concerns raised by Chen et al. (2018, 2023) that inferences can be incorrect when the residual from a regression model is used as the dependent variable. Specifically, our result is robust to adding assurance as an additional determinant in the carbon emissions model, including all determinants from the carbon emissions model as additional controls, and estimating quantile regressions with the level of emissions. See Tables IA–8

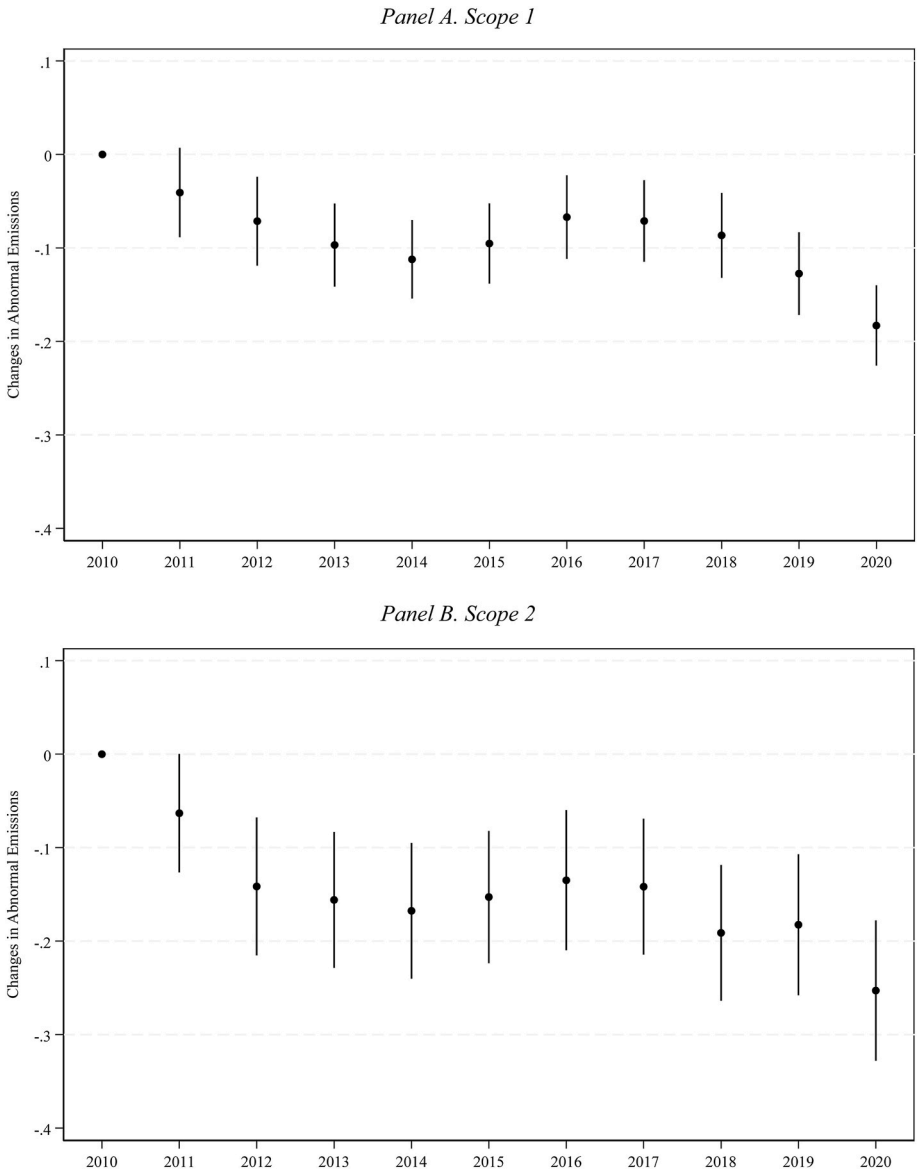


Fig. 2. Intertemporal Changes in Abnormal Emissions
This figure presents time trends of *Abnormal Emissions*. For each scope, *Abnormal Emissions* is regressed on calendar-year indicators, with 2010 omitted as the base year. Coefficient estimates for year indicators (dots) and 90 % confidence intervals (bars) are presented for each coefficient. Panels A and B report results for Scopes 1 and 2, respectively. All calculations are based on the full sample of data at the firm-year level.

association is driven by a latent variable, such as firm type. For example, a transparent firm simultaneously chooses assurance and a high-quality carbon accounting system that produces high-quality carbon emissions numbers. We address this concern in several ways. First, we use firm fixed effects to remove variation driven by firm type. As we show in Panel A of Table 4 columns (1) and (4), the result is robust to within-firm variation.

Second, we explore changes in *Abnormal Emissions* following first-time assurance adoptions. To do so, we create an indicator equal to one if an observation is in or after the first year with assurance for each scope. We regress *Abnormal Emissions* on this indicator, with either industry or firm fixed effects, akin to a staggered difference-in-differences design. As we show in columns (2), (3), (5), and (6), the results continue to indicate a negative association between *Abnormal Emissions* and *Assurance*. Comparing the coefficients in, e.g., columns (1) and (3), we find that *Post Assurance* appears to have a larger effect than *Assurance*, indicating that annual assurance might not be necessary to achieve the measurement benefits from verification.

Table 2
Abnormal Emissions and ESG Reporting Quality Measures

Independent Variable:	(1)	(2)	(3)	(4)	(5)	(6)
	Manager Uncertainty		*Refinitiv Disclosure Score*		*Bloomberg E Disclosure Score*	
	Dependent Variable: *Abnormal Emissions*					
Independent Variable	0.010***	0.013***	−0.249***	−0.103***	−0.460***	−0.341***
	(3.41)	(3.14)	(-6.19)	(-3.20)	(-5.98)	(-3.54)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Controls, Fixed Effects	No	No	No	No	No	No
Observations	2135	2038	3692	3542	1765	1696
R^2	0.008	0.009	0.026	0.004	0.031	0.016

This table presents results from regression analyses of model-estimated carbon accounting quality (*Abnormal Emissions*) on various perception-based measures of ESG reporting quality, including managers' perception of carbon accounting quality (*Manager Uncertainty*) in columns (1) and (2), Refinitiv ESG disclosure scores (*Refinitiv Disclosure Score*) in columns (3) and (4), and Bloomberg environmental-specific disclosure scores (*Bloomberg E Disclosure Score)* in columns (5) and (6). Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two–sided) at the 0.1, 0.05, and 0.01 levels, respectively.

Relatedly, we explore how assurance shapes carbon accounting quality in event time. We regress *Abnormal Emissions* on indicators for each year relative to the first year of assurance adoption. As shown in Fig. 3, in the pre-period, the coefficients are small, statistically insignificant, and exhibit no apparent trends for Scope 1 emissions. For Scope 2 emissions, all coefficients except for one are statistically insignificant in the pre-period, with a slight downward trend from $t = -2$. In the post-period, for both scopes, all coefficients are negative and statistically significant. Importantly, there is a significant "jump" in the coefficient magnitude in the first year of assurance (from $t = -1$ to $t = 0$). This trend is consistent with assurance improving carbon accounting quality, as opposed to an alternative "assurance readiness" explanation in which firms *choose* to obtain assurance when their carbon accounting quality is already high (or has recently improved) and assurance serves solely as a signaling device. Further, the coefficient magnitude becomes more negative over time, suggesting that firms continue to work with assurors to improve their carbon accounting quality.[16]

Third, we show that the association between carbon assurance and carbon accounting quality is scope-specific. To do so, we include assurance indicators for both Scope 1 and Scope 2 and study their associations with Scope 1 and Scope 2 abnormal emissions, feasible because there is variation in assurance across scopes. As we show in column (1) of Table 4 Panel B, when *Abnormal Emissions* is measured for Scope 1 emissions, the coefficient on *Scope 1 Assurance* has a larger economic magnitude and is more statistically significant relative to the coefficient on *Scope 2 Assurance*. Similarly, when *Abnormal Emissions* is measured for Scope 2 emissions in column (2), the coefficient on *Scope 2 Assurance* has a larger magnitude, a negative association, and significance, unlike the coefficient on *Scope 1 Assurance*.

Fourth, we show that assurance over emissions metrics does not statistically correlate with measures of financial reporting quality, including an indicator of whether a firm just meets or beats its earnings target (Burgstahler and Dichev, 1997) and abnormal accruals per McNichols (2002). If our main result is driven by overall firm transparency or financial reporting quality, we would expect a positive association between carbon assurance and financial reporting quality. However, as we show in columns (3)–(6), we do not find carbon emissions assurance to be correlated with measures of financial reporting quality, suggesting that our finding is unique to carbon reporting.

4.3. Cross-sectional analysis

4.3.1. Variation in the strength of carbon accounting system

Because assurance improves carbon accounting quality by helping firms identify issues in their carbon accounting systems, we expect our finding to be weaker for firms with robust systems already in place. To examine this prediction, we identify three scenarios. First, the U.S. Environmental Protection Agency's (EPA) Greenhouse Gas Reporting Program (GHGRP), established in 2010, requires facilities above a certain emissions threshold to report to the EPA and has been shown to affect emissions reporting (Tomar, 2023). EPA reporting is subject to electronic checks and manual reviews, so GHGRP firms likely have proper systems in place at least for reporting facilities, limiting the benefits of assurance. Second, responding to the CDP questionnaire can improve firms' carbon accounting systems. CDP reports that "7 out of 10 companies agree that disclosure through CDP helps [the companies] learn more about their

[16] This finding is consistent with anecdotal evidence in CDP data. Firms regularly discuss how assurors help them identify issues and subsequently improve their emissions reporting. For example, one company wrote that "[t]here is reliance on manual data transfer due to the variety of input sources. However, a 3rd-party certification firm assesses the database, data collection process, and reported data, and provides an assurance statement."

Table 3
Assurance and Carbon Accounting Quality
This table presents results from regressions of carbon accounting quality measures on assurance. Panel A reports the main result. Panel B reports results using two alternative measures of carbon accounting quality: firm perception of reporting uncertainty in Scopes 1 and 2 emissions and the number of days it takes for a firm to submit its questionnaire response to the CDP. *Abnormal Emissions, Manager Uncertainty*, and *Assurance* are all scope-specific, with the scope used indicated at the bottom of the table. *Filing Lag* varies only by firm-year pairs. Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two-sided) at the 0.1, 0.05, and 0.01 levels, respectively.

Panel A. Main Result

	(1)	(2)	(3)	(4)
Dependent Variable: *Abnormal Emissions*				
Assurance	−0.162***	−0.082***	−0.163***	−0.104***
	(-11.39)	(-5.27)	(-10.47)	(-6.57)
Controls:				
Size	–	−0.067***	–	−0.031***
		(-9.98)		(-4.73)
ROA	–	−0.055	–	−0.117
		(-0.51)		(-0.92)
Book-to-Market	–	−0.011	–	0.038*
		(-0.82)		(1.96)
Leverage	–	0.002	–	−0.001
		(1.45)		(-0.87)
Loss	–	0.026	–	0.027
		(0.95)		(0.88)
Sample	Scope 1	Scope 1	Scope 2	Scope 2
Fixed Effects	No	Ind, Year	No	Ind, Year
Observations	5395	5395	5202	5202
R^2	0.033	0.096	0.030	0.145

Panel B. Alternative Measures of Carbon Accounting Quality

	(1)	(2)	(3)	(4)
	Dependent Variable:			
	Manager Uncertainty		*Filing Lag*	
Assurance	−0.401**	−0.687***	−0.011*	−0.009*
	(-2.14)	(-3.71)	(-1.93)	(-1.69)
Sample	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	3305	3184	4733	4733
R^2	0.068	0.090	0.712	0.712

environmental impact and impact management," and CDP ratings reward higher-quality responses (Cohen et al., 2023b). Lastly, emissions reduction targets, typically set by the board of directors or management, signal a firm's commitment to sustainability and are associated with subsequent reductions in firm emissions (e.g., Dahlmann et al., 2019). The existence of emissions reduction targets likely translates into firms carefully tracking and, thus, being able to provide high-quality emissions information to stakeholders, which again likely limits the benefits of assurance.

To test these predictions, we estimate Eq. (2). We additionally include the partitioning variable—*CAS*—to control for the inherent difference in abnormal emissions across firms with EPA reporting, CDP disclosure, and scope-specific targets. Table 5 Panel A shows consistent evidence in all three settings. In columns (1) and (2), the main effect coefficient for firms subject to EPA reporting is negative and significant for Scope 1 but not Scope 2 emissions, indicating higher Scope 1 carbon accounting quality. This is a relevant difference between emissions scopes because facility emissions are Scope 1 by definition, so the EPA reporting requirements may have an effect mainly (a limited effect) on Scope 1 (Scope 2) emissions. We additionally find a lower effect of assurance for Scope 1 emissions of EPA reporting firms but similar improvements in carbon accounting quality for Scope 2 emissions for all firms. In columns (3) and (4), we show that the main effect coefficient is negative and significant for firms reporting to CDP for both Scope 1 and 2 emissions, consistent

Table 4
Addressing Alternative Explanations
This table presents results from analyses addressing potential alternative explanations. Panel A reports results using alternative fixed effects and research designs. Panel B reports results from falsification tests. *Scope 1 Assurance* and *Scope 2 Assurance* are the two separate scope-specific indicators for the assurance that apply to the scopes in the variable names. Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two-sided) at the 0.1, 0.05, and 0.01 levels, respectively.

Panel A. Alternative Research Designs

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: *Abnormal Emissions*						
Assurance	−0.032*	–	–	−0.049***	–	–
	(-1.94)			(-2.91)		
Post Assurance	–	−0.140***	−0.065**	–	−0.148***	−0.067**
		(-8.31)	(-2.39)		(-7.94)	(-2.25)
Sample	Scope 1	Scope 1	Scope 1	Scope 2	Scope 2	Scope 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Firm, Year	Ind, Year	Firm, Year	Firm, Year	Ind, Year	Firm, Year
Observations	5205	5395	5205	5017	5202	5017
R^2	0.367	0.108	0.368	0.421	0.155	0.421

Panel B. Falsification Tests

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent Variable:					
	Abnormal Emissions		Meet or Beat		Abnormal Accruals	
Scope 1 Assurance	−0.072**	0.020	−0.003	–	0.002	–
	(-2.44)	(0.30)	(-0.49)		(0.47)	
Scope 2 Assurance	−0.011	−0.123*	–	−0.005	–	0.005
	(-0.37)	(-1.82)		(-0.87)		(1.24)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5395	5202	5371	5182	4963	4744
R^2	0.096	0.145	0.141	0.144	0.173	0.175

with our argument above, and a reduced carbon accounting quality effect from assurance for both emissions scopes. In columns (5) and (6), we find a strong negative main effect, i.e., higher carbon accounting quality, for firms with targets and essentially no effect from assurance.[17] Together, the findings provide additional support that assurance improves carbon accounting quality, particularly for firms with *ex ante* weaker systems that are not strengthened by regulation or voluntary reporting regimes.

4.3.2. Variation in assurance thoroughness

If assurance improves carbon accounting quality, our main finding should be stronger when assurance is more thorough. We leverage detailed assurance data and develop three measures of assurance effort. Our first measure is the thoroughness of assurance within a scope—the contractual level of assurance. When firms hire an assurance provider, the engagement contract specifies whether the outcome of the assurance engagement is reasonable assurance (as with audits of annual financial statements) or limited assurance, which is less involved in terms of the work performed by the assuror (as with reviews of quarterly financial statements). Because reasonable assurance is much more thorough, we expect the association between assurance and *Abnormal Emissions* to be stronger for these engagements. Our second measure is whether a firm assures all three scopes of reported emissions. Covering all scopes requires the provider to review more data and investigate the carbon accounting system more thoroughly. In addition, because firms choose

[17] Importantly, the coefficient on *CAS* for emissions target is significantly negative, suggesting that firms with targets have smaller abnormal emissions. This finding mitigates a measurement concern that firms that actively decarbonize through divestments and efficiency improvements might behave differently than other firms in the sample. If this concern is valid, we expect decarbonization firms—which are more likely to adopt targets—to have higher abnormal emissions. In addition, because our emissions model relies on data from financial statements, if a firm's organizational boundary for carbon accounting is different from the boundary used to consolidate financial statements, then our abnormal emissions measure can be noisy. In Tables IA–9, we partition our sample by various proxies of organizational boundary complexity (e.g., magnitudes of equity investment, rental expense, non-controlling equity) and show that our finding exists in both subsamples. In addition, abnormal emissions do not differ statistically between the two samples.

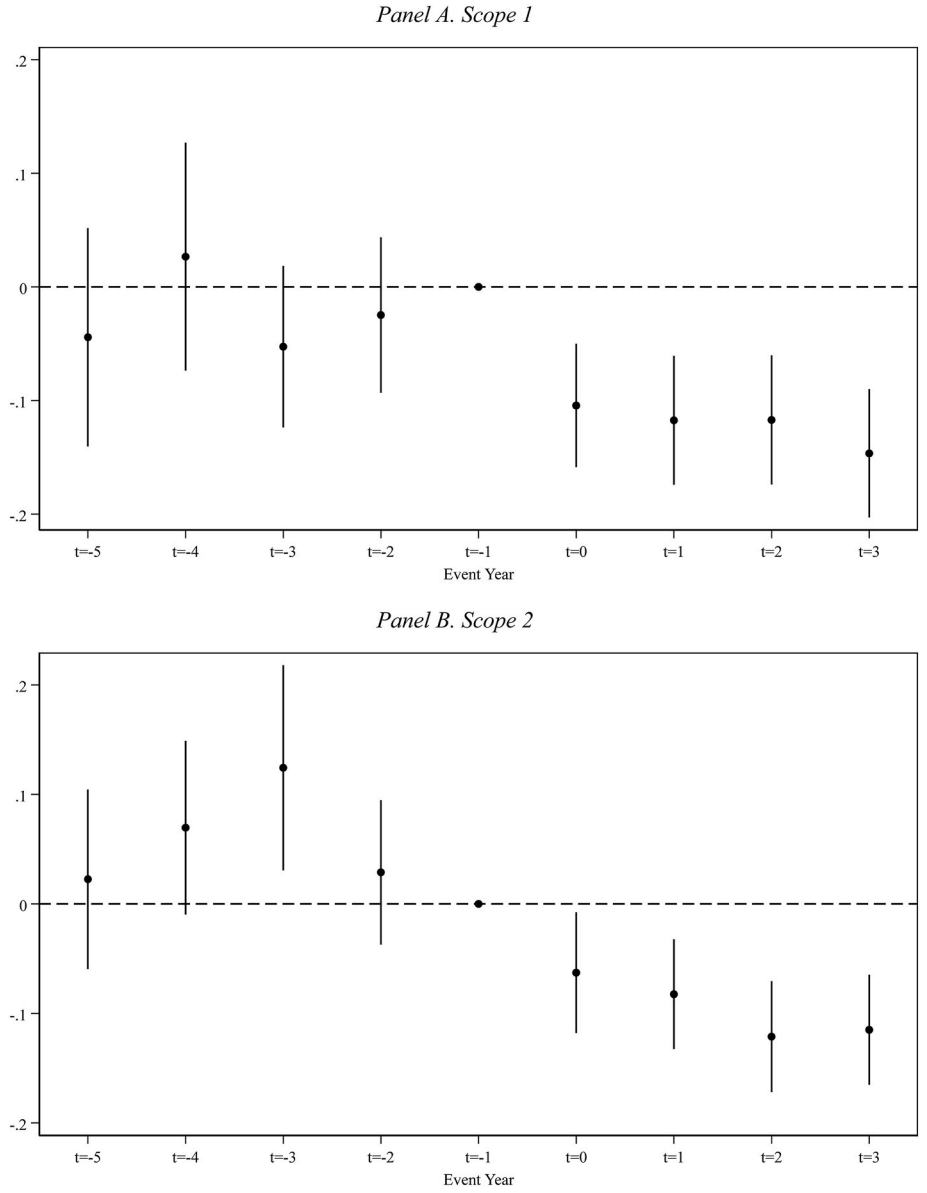


Fig. 3. Effect of First-Time Assurance on Abnormal Emissions in Event Time
This figure presents event-time analysis of the effect of first-time scope-specific assurance on *Abnormal Emissions* (y-axis). The first year with assurance is t = 0. All coefficients are benchmarked to the year prior to assurance adoption (t = −1). Coefficient estimates (dots) and 90 % confidence intervals (bars) are presented for each coefficient.

reporting boundaries that separate Scopes 1, 2, and 3 emissions, engaging an assuror for all three scopes of emissions allows the assuror to more accurately identify reporting boundaries and therefore reduce omissions and misclassifications. Our third measure combines these two dimensions and partitions *Assurance* into four non-overlapping cases: (1) limited assurance with not all three scopes assured, (2) limited assurance with all three scopes assured, (3) reasonable assurance with not all three scopes assured, and (4) reasonable assurance with all three scopes assured. We expect scenario (4) to have the highest assurance quality and (1) the lowest, with no strong priors about (2) and (3).

We report the findings in Table 5 Panel B, with odd and even columns reporting the results for Scopes 1 and 2, respectively.

Table 5
Cross-sectional Analysis
This table presents various cross-sectional analyses. Panel A shows whether the relation between assurance and carbon accounting quality varies by the presence of facilities covered by the EPA's GHGRP regime (columns 1 and 2), reporting to the CDP (columns 3 and 4), and emissions reduction targets (columns 5 and 6). Panel B shows whether the relation between assurance and carbon accounting quality varies by contracted assurance characteristics. Depending on the specification, the *Assurance* variable is redefined to indicate whether the assuror performs a limited level of scope-level assurance and/or assurance is provided over all three emissions scopes. Panel C shows whether the relation between assurance and carbon accounting quality varies by assuror type (columns 1 and 2) and assuror identity (columns 3 and 4). In columns 3 and 4, we separately list the Big 4 financial auditors (Deloitte, EY, KPMG, and PwC) as well as four major non-financial assurors (Apex, Bureau Veritas, ERM, and LRQA); all other assurance providers (mainly non-financial) fall into the *Other* category. The sample is limited to firms with emissions-specific assuror data. Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two-sided) at the 0.1, 0.05, and 0.01 levels, respectively.

Panel A. Variation in Carbon Accounting System Strength

	(1)	(2)	(3)	(4)	(5)	(6)
Carbon Accounting System (*CAS*) Measure	*EPA Coverage*		*CDP Reporting*		*Scope-Specific Emissions Target*	
Dependent Variable: *Abnormal Emissions*						
[A] *CAS = 0 × Assurance*	−0.101***	−0.093***	−0.119***	−0.182***	−0.096***	−0.127***
	(-5.06)	(-5.42)	(-4.66)	(-6.77)	(-5.20)	(-6.55)
[B] *CAS = 1 × Assurance*	−0.042**	−0.128***	−0.037**	−0.060***	−0.007	−0.015
	(-2.06)	(-4.38)	(-1.99)	(-3.01)	(-0.34)	(-0.72)
Main Effect:						
CAS	−0.058**	0.022	−0.098***	−0.089***	−0.129***	−0.139***
	(-2.42)	(0.74)	(-3.82)	(-3.39)	(-5.53)	(-5.58)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5395	5202	5395	5202	5395	5202
R^2	0.097	0.145	0.100	0.149	0.102	0.152
Pr([A]=[B]) p-value:	0.027	0.280	0.006	0.000	0.001	0.000

Panel B. Variation in Assurance Thoroughness

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: *Abnormal Emissions*						
[A] *Limited Assurance*	−0.080***	−0.101***	–	–	–	–
	(-4.89)	(-6.27)				
[B] *Reasonable Assurance*	−0.090***	−0.117***	–	–	–	–
	(-4.63)	(-5.32)				
[A] *Some Scopes Assured*	–	–	−0.076***	−0.088***	–	–
			(-4.04)	(-4.60)		
[B] *All Scopes Assured*	–	–	−0.085***	−0.110***	–	–
			(-5.06)	(-6.56)		
[1] *Limited + Some*	–	–	–	–	−0.072***	−0.088***
					(-3.60)	(-4.24)
[2] *Limited + All*	–	–	–	–	−0.083***	−0.087***
					(-3.01)	(-3.18)
[3] *Reasonable + Some*	–	–	–	–	−0.083***	−0.105***
					(-4.68)	(-6.23)
[4] *Reasonable + All*	–	–	–	–	−0.095***	−0.131***
					(-4.38)	(-5.10)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5395	5202	5395	5202	5395	5202
R^2	0.096	0.145	0.096	0.145	0.096	0.146
Pr([B]>[A]) p-value:	0.262	0.179	0.281	0.083	–	–
Pr([4]>[1]) p-value:	–	–	–	–	0.162	0.053

Panel C. Variation in Assurors

(*continued on next page*)

Table 5 (*continued*)

Panel C. Variation in Assurors				
	(1)	(2)	(3)	(4)
	(1)	(2)	(3)	(4)
Dependent Variable: *Abnormal Emissions*				
[A] *Non-financial Assuror*	−0.066***	−0.081***	–	–
	(-3.61)	(-3.38)		
Apex	–	–	−0.073**	−0.081**
			(-1.98)	(-2.50)
Bureau Veritas	–	–	−0.076**	−0.087***
			(-2.54)	(-3.22)
ERM	–	–	−0.076**	−0.084***
			(-2.56)	(-3.19)
LRQA	–	–	−0.078*	−0.167***
			(-1.67)	(-3.06)
Other	–	–	−0.062***	−0.073***
			(-3.29)	(-2.83)
[B] *Financial Auditor*	−0.061*	−0.057	–	–
	(-1.96)	(-1.59)		
Deloitte	–	–	−0.087**	−0.089
			(-2.33)	(-1.45)
EY	–	–	−0.101***	−0.002
			(-3.05)	(-0.05)
KPMG	–	–	−0.094*	−0.165**
			(-1.84)	(-2.24)
PwC	–	–	0.136	−0.048
			(1.06)	(-1.00)
Sample	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	2282	2169	2282	2169
R^2	0.059	0.219	0.062	0.221
Pr([A]=[B]) p-value:	0.857	0.428	–	–

Columns (1) and (2) focus on the contractual level of assurance. For both scopes, the coefficient on *Reasonable Assurance* is larger, meaning that relative to limited assurance, reasonable assurance further reduces the level of abnormal emissions, though the difference is not significant at conventional levels.[18] In columns (3) and (4), we examine the pervasiveness of assurance. Consistent with more thorough assurance resulting in higher reporting quality, firms that assure all three scopes have lower levels of abnormal emissions. Notably, the difference in coefficients is statistically significant at conventional levels for Scope 2 emissions. In columns (5) and (6), we find the coefficient magnitude of *Assurance* to be the largest for firms with reasonable assurance and all three scopes assured, and smallest for limited assurance and some scopes assured. In terms of the magnitude difference, having reasonable assurance and all three scopes assured results in a 32 % and 49 % increase in carbon accounting quality relative to having limited assurance and only some scopes assured. In aggregate, these analyses provide evidence broadly consistent with thorough assurance improving carbon accounting quality.

4.3.3. Variation in assurors

Our next analysis explores whether our main result is heterogeneous across financial auditors and non-financial assurors.[19] On the one hand, experimental studies show that analysts trust financial auditors more and *perceive* assurance conducted by financial auditors to be of higher quality (Pflugrath et al., 2011). On the other hand, auditing carbon emissions requires expertise in environmental science, and non-financial assurors tend to have such human capital. Therefore, we do not have directional expectations. As we show in columns (1) and (2) of Table 5 Panel C, relative to financial auditors, the coefficient for assurance conducted by non-financial assurors

[18] Anecdotally, when a firm first obtains assurance, the reporting firm and the assuror typically start with limited assurance and sometimes set the goal to eventually converge to reasonable assurance, where improved systems and reporting quality enable the assuror to provide reasonable assurance. This upward trajectory implies that within the category of limited assurance there is likely heterogeneity in reporting quality. The pooling of reporting types within limited assurance could explain why the magnitude difference is not significant at conventional levels.

[19] Because CDP does not track assuror identity, this analysis is limited to the firms covered in Gipper et al. (2025), i.e., firms that issued ESG reports and explicitly mentioned the assuror identity.

is statistically significant and economically larger for both Scope 1 and Scope 2 emissions, although the Wald test does not show statistical differences between the two groups at conventional levels. Moreover, assurors may have different effects on carbon accounting quality apart from a simple classification as either a financial auditor or a non-financial assuror. To explore this possibility, we partition the *Assurance* indicator into nine granular indicators, including separate indicators for the four largest non-financial assurors and the Big 4 financial auditors, as well as an "other" category for the remaining (mainly non-financial) assurors. In columns (3) and (4), we find that the coefficients for all non-financial assurors are significant at conventional levels and significant results for three of the Big 4 auditors, suggesting that the slightly weaker coefficient estimate in column (1) is likely driven by PwC. For Scope 2 emissions, only assurance with KPMG has a statistically significant correlation with *Abnormal Emissions*. The results indicate potential heterogeneity within financial auditors, though also potentially noisy estimates due to auditors' low market share, something which might evolve as financial auditors build this practice.

4.4. Mechanism

Our next set of tests focuses on *how* assurance improves carbon accounting quality. As discussed in Sections 2.2 and 2.3, this process likely involves assurors identifying and fixing various implementation issues. If so, we expect assurance to be associated with various *indicators* of the carbon accounting system. To test this hypothesis, we turn to CDP data and develop measures based on firms' responses to CDP questionnaires.[20]

We first focus on indicators that are inputs to generating carbon accounting numbers. In the assurance process, assurors will likely identify various issues related to carbon accounting, including using incorrect conversion factors and failing to account for energy use in a facility, among others. Therefore, we expect *Assurance* to be positively associated with the probability of issues being identified (*Issues Identified*), and consequently, negatively associated with the number of emissions sources being omitted when reporting emissions (*Omissions*). As we show in columns (1) through (4) of Table 6, assurance increases issues identified and decreases omissions. Interestingly, the magnitude of the association between *Assurance* and *Omissions* is larger for Scope 2 emissions. One possibility is that emission sources for Scope 1 are easier to comprehensively identify in the first place, limiting the incremental role of assurance.

Once implementation issues in the carbon accounting system are identified, assurors would communicate the issues to the reporting firm. To the extent firms take remedial actions, we would expect historically reported numbers to be revised. For example, if an assuror identified that a reporting firm's emissions number fails to account for emissions from an overseas facility, the firm could recalculate and update the number. More formally, the GHG Protocol explicitly states that revisions are required when (1) there are significant changes in calculation methodology, which typically result from improved emission factors and/or improved activity data, and (2) significant errors or smaller errors that are collectively significant are discovered.[21] Consistent with our prediction and prior literature, the results in columns (5) and (6) show that assurance is associated with an increased probability of firms revising their historical emissions figures.

Taken together, the results in Tables 3 and 6 describe how assurance improves carbon accounting quality. First, assurors identify issues and communicate them to the reporting firm, and there is a decline in reporting omissions, consistent with an improved and more complete carbon accounting system. Then, firms take additional remedial actions and improve their carbon reporting. As a result, managers believe that their reported emissions figures are more precise, and firms can assemble and file emissions information with CDP in a timelier manner.[22]

4.5. Shock to ESG assurance: E.U.'s Non-Financial Reporting Directive

Thus far, our analysis focuses on voluntary assurance. In the next analysis, we utilize the implementation of the E.U.'s Non-Financial Reporting Directive 2014/95/EU (NFRD) to further examine how assurance shapes carbon accounting quality in a mandatory assurance setting. Adopted in 2014 and effective in most E.U. member states beginning in 2018, NFRD requires public-interest entities (such as publicly listed companies, banks, or insurance companies) exceeding 500 employees on average to report on a variety of ESG issues, including environmental matters. If a firm considers climate as a material issue, the firm is required to

[20] For Table 6, sample size varies by dependent variable. This is because CDP questionnaires vary by year and some questions are not consistently asked. In addition, although CDP tracks when firms submit their responses, this information is not available to us for some years in our sample period.

[21] This argument is similar to financial auditors identifying internal control weaknesses after SOX, resulting in more frequent restatements and disclosure of control weaknesses (e.g., Ge and McVay, 2005; Doyle et al., 2007; Rice and Weber, 2012). However, in emissions reporting, measurement standards and assurance practices are not as consistent and well understood as in financial reporting, leading to higher revision rates than financial statement restatements (e.g., Michelon et al., 2019). We use different terminology, "revisions," reflecting that the practices are different, such as revising emission numbers from prior years for methodological changes—a practice that is specifically disallowed in financial reporting. Moreover, assurors may signal quality through emissions revisions (Michelon et al., 2019), but undiscovered financial misstatements later revealed through a restatement are a signal of low audit quality (DeFond and Zhang, 2014).

[22] The fact that assurance increases the number of issues identified helps us rule out the alternative story that firms assure their emission numbers when their carbon accounting quality is high or improving. If the alternative story is true, we should observe a decrease in the number of issues identified with the carbon accounting system.

Table 6
Mechanism: Assurance and Indicators of the Carbon Accounting System
This table presents results from an analysis of the association between assurance and three indicators of the carbon accounting system: the number of issues identified (*Issues Identified*), the probability that at least one emissions source is omitted (*Omissions*), and whether historical GHG emissions numbers are updated (*Revisions*). Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two-sided) at the 0.1, 0.05, and 0.01 levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent Variable:					
	Issues Identified		*Omissions*		*Revisions*	
Assurance	0.248***	0.291***	−0.059**	−0.119***	0.015*	0.020***
	(3.01)	(3.59)	(-2.21)	(-4.56)	(1.88)	(3.13)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Ind, Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3517	3454	5215	5215	5215	5215
R^2	0.080	0.067	0.040	0.046	0.053	0.035

disclose climate information. As part of NFRD, all E.U. countries have requirements for the statutory auditor to check whether non-financial information has been provided. On top of that, 11 countries have additional requirements for the auditor to check the consistency of non-financial information with the financial statements. Finally, France, Italy, and Spain require mandatory independent assurance over the non-financial information from this directive.[23]

This analysis of mandatory assurance in the E.U. serves two purposes, both complementary to the voluntary assurance that we examine in the U.S. First, because country-level implementation of NFRD is unlikely the result of any one firm's choice to obtain assurance, variation in assurance is less likely to be caused by other, endogenous reasons at the firm level. For example, reverse causality—i.e., carbon accounting quality improvements lead firms to obtain assurance—is not likely to explain the findings in this setting. Second, studying mandatory carbon assurance itself is relevant. Prior studies suggest that mandatory financial statement audits for firms have a non-obvious impact. These audits may primarily address specific agency issues—like the separation of ownership and control—and be an economic drag when the bulk of firms either do not have such issues or they can be addressed with other mechanisms (e.g., Breuer, 2021). Although the incentives to voluntarily obtain carbon assurance and to comply with carbon assurance mandates are inherently different, we suspect that both voluntary and mandatory assurance can improve reporting quality because carbon accounting is still in its infancy, meaning that firms likely realize measurement benefits regardless of the incentive to obtain assurance. As carbon reporting and assurance mature, the effects of mandates might change, ultimately looking more like financial statement audits.

We leverage this institutional setting and examine whether mandatory assurance over non-financial information, which likely includes GHG emissions for many firms, increases carbon accounting quality. To do so, we exploit within-E.U. variation in assurance and estimate difference-in-differences regressions as follows:

$$CAQ_{i,t,s} = \beta_1\, Mandatory \times Post + \text{Controls} + \text{Firm, Year FE} + \varepsilon_{i,t,s} \tag{4}$$

in Eq. (4), the dependent variable is *Abnormal Emissions* for firm *i*, Scope *s*, in year *t*. Because our baseline *Abnormal Emissions* measure is estimated for U.S. companies only, we make two modifications to our carbon emissions model to generate this variable for non-U.S. companies. First, we include country fixed effects, recognizing the heterogeneity in production technologies across countries. For example, a firm from a less developed country might use old production technologies that generate more emissions than a firm from an advanced economy with state-of-the-art equipment. Second, because our measure of production centrality (the number of states and countries with production facilities) is unavailable for non-U.S. countries, we drop this determinant. We estimate *Abnormal Emissions* for countries with available disclosed emissions data from Trucost and firm fundamentals from Compustat Global.[24] *Mandatory* is an indicator variable measured at the country level and equals 1 if a firm is in France, Italy, or Spain, and 0 otherwise. *Post* equals 1 if the observation is after NFRD implementation, and 0 otherwise.

We use the same set of control variables. In addition, we include firm and year fixed effects, which subsume *Mandatory* and *Post*, respectively. The use of firm fixed effects is particularly important in this setting. Given that NFRD is not a regulation specific to

[23] In Figure IA-1, we confirm that firms in these countries have relatively higher assurance rates beginning in 2018.

[24] In Tables IA–10, we describe the composition of the non-U.S. sample, present statistics, and show that assurance is correlated with reduced *Abnormal Emissions*, generalizing our baseline U.S. finding to 42 non-U.S. countries. We construct firm-level controls in reporting currency; variables are either ratios (which remove scale differences) or logarithms (where scale differences are collinear with fixed effects). Untabulated tests that convert currencies to USD show highly comparable results with respect to magnitudes and significance.

greenhouse gas emissions, firm fixed effects allow us to abstract away from firm-invariant confounders, e.g., the types of non-financial information disclosed. We use an event window from 2015 to 2019, three years before (to assess trends) and two years after NFRD implementation. Lengthening the post period by one year to balance the pre and post periods does not change our inferences (untabulated). For this analysis, we consider two sets of control firms. The first set consists of all firms in E.U. countries without an assurance mandate, and the second set consists of firms in E.U. countries that require auditor consistency checks. Our coefficient of interest is on *Mandatory* × *Post* (i.e., β_1), which we predict to be negative. As we show in columns (1)–(4) of Table 7, mandatory assurance improves the reporting quality for Scope 1 emissions by roughly 13 % within-firm. For Scope 2 emissions, firms in countries with the assurance mandate still experience lower abnormal emissions post-NFRD; however, the magnitudes are smaller at around 9 %. These effects are larger point estimates than those using the U.S. sample and firm fixed effects in Table 4.

To provide additional credibility, we perform a falsification test. Recall that 11 countries require auditors to check the consistency of non-financial information with the financial statements but do not require mandatory assurance. If our documented effect is not assurance-specific, then we would expect a similar result for those 11 countries. In columns (5) and (6), we perform a similar DiD analysis. In this falsification test, the treatment group includes firms from the 11 countries and the control group includes firms from the remaining E.U. member states, excluding France, Italy, and Spain. Consistent with our conjecture, we find no evidence that "consistency checks" improve carbon accounting quality. Together, these findings support our inference that assurance shapes carbon accounting quality.[25]

5. Conclusion

This study examines the role of third-party assurance in shaping carbon accounting quality. We propose a conceptual framework of carbon accounting quality and introduce a scope-specific measure inspired by abnormal accruals models. We document a positive association between assurance and carbon accounting quality for both U.S. and non-U.S. countries. This relation is stronger when firms have *ex ante* weak carbon accounting systems and when assurance is more thorough. We also document how assurance improves carbon accounting quality: first, assurors identify issues in the carbon accounting system and communicate them to the firm; subsequently, firms take remedial actions, resulting in updated disclosures. Using the implementation of NFRD in the E.U. as an alternative setting, we show that mandatory assurance improves carbon accounting quality.

Our study contributes to the ESG reporting quality literature by introducing a novel measure of emissions reporting quality. Notably, this measure can be derived for any firm disclosing emissions data (though with a sufficient mass of disclosing peers) and is scope-specific. We believe this measure could be useful for academic researchers and stakeholders attempting to assess carbon disclosure quality. Our study also contributes to the ESG assurance literature by documenting the economic effects of assurance. Gipper et al. (2025) document significant heterogeneity in ESG assurance and the pervasive use of limited assurance in ESG reporting, casting doubt on the usefulness of ESG assurance. On the contrary, our findings suggest that even limited assurance can shape carbon accounting quality.

We acknowledge that our regression-based measure of abnormal emissions is subject to model misspecification and measurement error. At least starting from Jones (1991), decades of research have refined earnings quality measures, and the literature is still expanding (Dechow et al., 1995; Dechow and Dichev, 2002; McNichols, 2002; Kothari et al., 2005; Breuer and Schütt, 2023). Similarly, we do not believe that our study has developed a definitive measure of carbon accounting quality, but we believe that our approach provides a useful starting point. Future research could either build on our models or devise alternative measures to further corroborate inferences across measures and compare different measures' strengths and weaknesses. Regarding assurance, our study does not establish causality, but our evidence is consistent with assurance driving improvements in carbon accounting quality. Given the looming assurance mandates in regulations such as California SB 253 and E.U. Corporate Sustainability Reporting Directive, we believe this could be another fruitful area for future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[25] We perform additional tests to strengthen the credibility of this analysis. First, prior to NFRD, some E.U. member states had country-specific carbon-related mandates, e.g., France (Bourveau et al., 2025b), or passed laws implementing NFRD at different specific dates around the common 2018 phase-in, e.g., Italy (Balluchi et al., 2020). In Tables IA–11, we drop treatment countries one at a time and find that lower *Abnormal Emissions* post-NFRD is not specific to any single country with the assurance mandate. Second, we plot out the NFRD effect in event time to assess the parallel trends assumption. As we show in Figure IA-1, we not only find no pre-trends prior to NFRD implementation, but also no dynamic effect for our falsification analysis from columns (5) and (6). Third, we perform an analysis to assess the robustness of the parallel trends assumption, allowing post period deviations from parallel trends to be between a 0 and 0.5 fraction of the largest deviation in the pre period (Rambachan and Roth, 2023). This test (untabulated) shows that the 90 % confidence intervals for the effect exclude zero for fractions $\leq$0.2 and only begin to include zero at fractions $\geq$0.3, indicating our results are robust to moderate violations of parallel trends. Lastly, we partition the DiD coefficient by examining the effect of the mandate separately for firms with and without voluntary assurance in the pre-NFRD years. As we show in Tables IA–12, the result is concentrated in firms without voluntary assurance, suggesting that the documented effect more likely comes from firms getting assurance as opposed to the assurance mandate further improving the quality of existing voluntary assurance.

Table 7
NFRD and Carbon Accounting Quality
This table presents results from analyses of the effect of NFRD on abnormal emissions. Firms in three countries—France, Italy, and Spain—became subject to NFRD with assurance due to corresponding adoption laws passed in each country that included an assurance component for non-financial information, i.e., Order No. 2017-1180 for French firms, Legislative Decree 254/2016 for Italian firms, and Law 11/2018 for Spanish firms. In columns (1) and (2), the sample is all European Union (E.U.) member countries for which we have data, plus Norway and the United Kingdom. In columns (3) and (4), the sample is E.U. countries whose sustainability disclosures are subject to mandatory assurance (France, Italy, and Spain), as well as E.U. member countries (plus Norway and the U.K.) for which we have data and whose sustainability disclosures are subject to auditor consistency checks (Austria, Belgium, Denmark, Finland, France, Germany, Netherlands, Norway, Slovenia, Sweden, and the U.K.). In columns (5) and (6), the sample is all E.U. member countries (plus Norway and the U.K.) for which we have data, excluding France, Italy, and Spain. In all columns, we restrict our sample to data from fiscal years 2015 through 2019. Definitions of all variables are as described in Appendix A. t–statistics appear in parentheses and are clustered by firm. *, **, *** indicate statistical significance (two-sided) at the 0.1, 0.05, and 0.01 levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: *Abnormal Emissions*						
Mandatory × *Post*	−0.133***	−0.092**	−0.136***	−0.093**	–	–
	(-3.48)	(-2.28)	(-3.52)	(-2.28)		
Falsification × *Post*	–	–	–	–	0.028	0.016
					(0.50)	(0.31)
Sample	Scope 1	Scope 2	Scope 1	Scope 2	Scope 1	Scope 2
Firms	All E.U. Countries		E.U. Countries with Mandatory Assurance or Auditor Consistency Checks		Excluding E.U. Countries with Mandatory Assurance	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm, Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3407	3290	3154	3092	2587	2467
R^2	0.400	0.450	0.401	0.439	0.391	0.466

Appendix A. Variable Definitions

This table presents variable definitions for our empirical tests.

Variable	Definition
Abnormal Accruals	A measure of firms' abnormal accruals per McNichols (2002).
Abnormal Emissions	The absolute value of the residual term from a regression of the natural logarithm of scope-level emissions (from Trucost) on its determinants. See Section 3.2 for methodological details. We also compute a version of this variable for non-U.S. firms. See Section 4.5 for methodological details.
Emissions Level	The absolute value of scope-level emissions (from Trucost). The unit of measurement is thousand tons of CO_2e.
All Scopes Assured	An indicator variable equal to one if assurance (from CDP and/or GRS data) was provided over all three emissions Scopes in a given firm-year, and zero otherwise.
Assurance	A scope-specific indicator variable equal to one if either GRS data or CDP indicates that a firm obtains assurance over emissions for a particular emissions Scope in a year, and zero otherwise.
Bloomberg E Disclosure Score	A proprietary score from Bloomberg based on the extent of a company's environmental disclosure. The score ranges from 0.1 (the minimum amount of environmental disclosure) to 100 (disclosure of every data point collected by Bloomberg related to the environmental component of ESG). We divide scores by 100 in regression analyses to facilitate comparisons in our models.
Book-to-Market	The ratio of the book value of equity to the market value of equity, measured at the fiscal year-end, from Compustat.
CDP Reporting	An indicator variable equal to one if a firm reports to CDP in a given firm-year, and zero otherwise.
EPA Coverage	An indicator variable equal to one if the firm has facilities covered by the EPA's Greenhouse Gas Reporting Program (GHGRP) regime in a given firm-year, and zero otherwise.
Falsification	An indicator variable equal to one if a firm is headquartered in one of the eleven countries that require auditors to check the consistency of non-financial information with the financial statements, and zero otherwise.
Financial Auditor	An indicator variable equal to one if the assurance provider type (according to GRS data) is a traditional financial auditor, and zero otherwise. In Table 5, we separately examine the identity of the Big 4 financial auditors: Deloitte, EY, KPMG, and PwC.
Filing Lag	The natural logarithm of the difference between the date a firm submits its response to the CDP questionnaire and March 1, an approximation for the annual CDP questionnaire release date.
Issues Identified	A scope-specific variable from CDP that counts the number of issues management identified with the carbon reporting system used to generate emissions.
Leverage	The ratio of total liabilities to total equity, measured at the fiscal year-end, from Compustat.
Limited Assurance	An indicator variable equal to one if the contracted level of assurance is limited according to CDP and/or GRS data, and zero otherwise. *Limited Assurance* and *Reasonable Assurance* span *Assurance*.
Loss	An indicator variable equal to one when basic earnings per share excluding extraordinary items from Compustat is less than zero, and zero otherwise.
Mandatory	An indicator variable equal to one if a firm is headquartered in France, Italy, or Spain, and zero otherwise.

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Variable	Definition
Manager Uncertainty	A scope-specific decile rank of management's estimate of the level of uncertainty of the firm's total emissions. The question appeared in some years' CDP questionnaires, usually in Section 8 under "Data Accuracy."
Meet or Beat	An indicator variable equal to one if income before extraordinary items is between 0 and 0.5 % of total assets, and zero otherwise.
Non-financial Assuror	An indicator variable equal to one if the assurance provider type (according to GRS data) is not a traditional financial auditor, and zero otherwise. In Table 5, we separately examine the identity of four major non-financial assurors: Apex, Bureau Veritas, ERM, and LRQA.
Omissions	A scope-specific variable that counts the number of sources of emissions which are not included in the number reported to CDP. This question appeared in some years' CDP questionnaires, usually in Section 8 under "Scope 1 and 2 Emissions Data."
Reasonable Assurance	An indicator variable equal to one if the contracted level of assurance is reasonable according to CDP and/or GRS data, and zero otherwise. *Reasonable Assurance* and *Limited Assurance* span *Assurance*.
Post	An indicator variable equal to one if the firm-year observation occurs after the implementation of the EU Non-Financial Reporting Directive (NFRD) in 2018, and zero otherwise.
Post Assurance	A scope-specific indicator variable equal to one if an observation is in or after the first year with assurance, and zero otherwise.
Refinitiv Disclosure Score	A proprietary score from Refinitiv of a company's ESG disclosures. The score ranges from 0 to 100 and is determined by Refinitiv's relative scoring algorithm. We divide scores by 100 in regression analyses to facilitate comparisons in our models.
Revisions	A scope-specific indicator variable from CDP equal to one if a firm revised its historical emissions number(s), and zero otherwise. We proxy for the revision based on year-on-year comparisons of CDP responses for reported, historical emissions.
ROA	Earnings before extraordinary items divided by average total assets, from Compustat.
Scope-Specific Emissions Target	An indicator variable equal to one if a firm has a scope-specific emissions target in a given year that it reports to CDP, and zero otherwise.
Size	The natural logarithm of the total assets measured at fiscal year-end, from Compustat.
Some Scopes Assured	An indicator variable equal to one if assurance (from CDP and/or GRS data) was not provided over all three emissions Scopes in a given firm-year, and zero otherwise.

Appendix B. Issues in the Carbon Accounting System

This appendix discusses issues in firms' carbon accounting systems and provides examples of firm-provided anecdotes from CDP responses regarding these issues. Responding firms select the "Explanation" categories.

Explanation	Example
Data Management	
Unlike financial accounting, where firms use established accounting software to track transactions and produce financial statements, the use of carbon accounting software is less common. Many firms track and calculate emissions using Excel spreadsheets. Even within a firm, establishments record data using different Excel templates with different data granularity. As a result, errors could arise from data entry, aggregation, and incorrect Excel formulas, among others. Data management issues could also arise when firms choose emissions conversion factors, such as using incorrect or outdated factors.	"Data management system is a combination of automated and some manual, Excel sheet-based calculations. Hence typographical errors, data handling errors are unavoidable, though we have ensured minimum error by our multi-level assurance check." "We are currently using Excel data sheets to collect our data and are in the process of integrating our GHG emissions data into our financial data reporting system." "The possibility of a human error occurring when entering data or using a calculation formula cannot be eliminated, because calculations are done using Excel instead of a calculation tool." "Manual errors in reading meters, reading from invoices, transferring data onto Excel spreadsheets etc. although these are reviewed at a number of levels." "Errors could occur when transferring data from supplier invoices, extrapolation of data within calculation models, and rounding errors."
Data Gaps	
Some firms do not have the proper information system in place to gather emissions data from certain sources. For example, a firm may not receive natural gas bills for leased facilities, and if the firm does not actively track down such information from the landlords, data gaps would exist. As another example, multinational firms may lack emissions data from certain overseas offices and establishments. Instead of estimating by making assumptions, some firms simply omit such offices and establishments from their reporting.	"Because we are a tenant in multi-tenant buildings we do not have direct control of or access to energy usage data for our facilities and, especially within the U.S. where we have the greatest concentration of facilities, our spaces are not separately submetered for utilities." "As we are a global business and market-based reporting is a new level of detail, many electricity providers in many countries are not familiar with conversion factors or what they mean. It is therefore difficult to get these. Where they are unavailable, we follow the GRI Reporting hierarchy for market-based factors, meaning that the reported figures may not be as accurate as they should as a result of the data gaps."
Assumptions and Extrapolation	
When data gaps exist, firms may either ignore data gaps (which leads to underreporting) or make assumptions and extrapolate. For example, when exact metered data for leased space is not available, a firm could extract a pro-rata share. However, some landlords do not have complete data, have inconsistent data, or are unresponsive to firm requests. In those cases, firms need to make further assumptions, such as using kWh electricity or natural gas per square foot averages in the region to extrapolate. In some cases, firms simply use the number from the previous reporting period.	"Data gathered was for entire buildings and then Capital Power's portion was determined based on square footage." "Where we do not have direct access to the data we rely on the building landlord to provide total building energy usage for the building, which we then prorate for our applicable portion of the total building space. When we are not able to obtain data from a landlord we must estimate energy usage using published energy intensity factors appropriate for each region." "This location shares utilities with a separate company. We split utilities by floor space. We claim 67 % of the total utility supplied to the facility."

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Explanation	Example
	"71 % of scope 1 and 2 emissions come from the energy bought for real estate investment activities and for corporate use. These emissions are calculated based on invoices collected for all the reporting scope. The rest of scope 1 and 2 emissions come from business travel done by employees. These emissions are calculated based on invoices collected or on estimated kilometers done. This can lead to higher uncertainty compared to data read on invoices."
Sampling Some firms do not measure Scopes 1 and 2 emissions for all sources; instead, they obtain some samples and extrapolate. Similarly, sampling is a common technique to generate data for Scope 3 emissions. Because the quantity of data that needs to be collected is large, a firm may find it impossible to collect data from each activity in a Scope 3 category. In those cases, a firm may use sampling to extrapolate data from a representative sample of activities within the category. For example, if a company wants to collect data on employee commuting (Category 7 of Scope 3 emissions), instead of collecting data for every single employee, the company may sample a subset of employees and extrapolate. Improper sampling techniques could introduce significant measurement errors.	"Calculation based on electric bill for a sample 3 month period." "To calculate the standard CO2 emissions produced at the company's work sites, we took a sample that was equivalent to 20 %–30 % of the completed construction amount. We calculate the total CO2 emissions amount based on the standard CO2 emissions calculated based on that sample."

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